PRACTICE QUESTIONS (SOLUTION)

By Pin2

MTH632

Lecture No. 1

Q1: Find the sum of two complex numbers, $z = 2 + i$, $w = -3 + 9i$.

Solution: $z + w = (2 + i) + (-3 + 9i)$

$= 2 + (-3) + i + 9i$

$= -1 + 10i$

Q2: Find the difference of two complex numbers, $z_1 = -5 + i$, $z_2 = -3 + 9i$.

Solution: $z_1 - z_2 = (-5 + i) - (-3 + 9i)$

$= -5 - (-3) + i - 9i$

$= -5 + 3 + i - 9i$

$= -2 - 8i$

Q3: Find the product of two complex numbers, $z_1 = -3 + 2i$, $z_2 = -1 + 11i$.

Solution: $z_1 z_2 = (-3 + 2i)(-1 + 11i)$

$= 3 - 33i - 2i + 22i^2$

$= 3 - 33i - 2i + 22(-1)$

$= -19 - 35i$

Q4: Divide of two complex numbers, $z_1 = -8 + i$, $z_2 = -3 + 8i$.

Solution:

z_1 棧 8棧+ 棧 z_2 棧8 + i

$$\frac{z_1}{z_2}=\frac{8+i}{3+8i}$$

$$=\frac{8+i}{3+8i}\times\frac{3-8i}{3-8i}$$

$$=\frac{24+64i-3i-8i^2}{(3)^2-(8i)^2}$$

$$=\frac{24+64i-3i-8(-1)}{9-64(-1)}$$

$$=\frac{24+64i-3i+8}{9-64(-1)}$$

$$=\frac{32+61i}{73}$$

Q5: Simplify $(\sqrt{3}+i\ 3)-i(5-\sqrt{3}i)$

Solution:

$$\begin{aligned}&(\sqrt{3}+i\ 3)-i(5-\sqrt{3}i)\\ =\ &\sqrt{3}+3i-5i+\sqrt{3}i^2\\ =\ &\sqrt{3}-2i+\sqrt{3}(-1)\\ =\ &\sqrt{3}-2i-\sqrt{3}\\ =\ &\sqrt{3}-\sqrt{3}-2i\\ =\ &0-2i\\ =\ &-2i\end{aligned}$$

Q6: Find *Re(z)* and *Im(z)* complex numbers, $z = -3 + 9i$.

Solution:

$Re(z) = -3$ and $Im(z)=9$

PRACTICE QUESTIONS

By Pin2

MTH632

Lecture No. 2

Modules (5 – 7)

Q1: Verify commutative law with respect to multiplication for complex numbers $z_1 = 2 + i$, $z_2 = -3 + 9i$.

Solution: The commutative law is $z_1 + z_2 = z_2 + z_1$

Consider $z_1 + z_2 = (2 + i) + (-3 + 9i)$

$= 2 - 3 + i + 9i$

$= -1 + 10i$

Now $z_2 + z_1 = (-3 + 9i) + (2 + i)$

$= -3 + 2 + 9i + i$

$= -1 + 10i$

Hence $z_1 + z_2 = z_2 + z_1$

Q2: Verify the associative law with respect to multiplication for complex numbers $z_1 = 2 + i$, $z_2 = -3 + 9i$ and $z_3 = -8 + i$.

Solution: The associative law is $(z_1 z_2) z_3 = z_1 (z_2 z_3)$

Consider $z_1 z_2 = (2 + i)(-3 + 9i)$

$= 2(-3) + 2(9i) + i(-3) + (i)(9i)$

$= -6 + 18i - 3i + 9i^2$

$= -6 + 18i - 3i + 9(-1)$

$= -6 - 9 + 18i - 3i$

$= -12 + 15i$

Now $(z_1 z_2) z_3 = (-12 + 15i)(-8 + i)$

$= -12(-8) + (-12(i)) + 15i(-8) + (15i)(i)$

$$= 96 - 12i - 120i + 15i^2$$

$$= 96 - 12i - 120i + 15(-1)$$

$$= 81 - 132i$$

Similarly, you can find $z_1(z_2 z_3)$.

Q3: Find the multiplicative inverse z^{-1} of the complex numbers $z = -5 + i$.

Solution:

$$z^{-1} = \frac{1}{z} = \frac{1}{-5+1} = \frac{-5}{(-5)^2+1^2} - i\frac{1}{(-5)^2+1^2} \quad \text{by formula}$$

$$= \frac{-5}{25+1} - i\frac{1}{25+1}$$

$$= \frac{-5}{26} - i\frac{1}{26}$$

Q4: Write down the Distributive law of multiplication over addition.

Solution: The Distributive law is $z_1(z_2 + z_3) = z_1z_2 + z_1z_3$

Q5: Let $z_1 = -8 + i$, $z_2 = -3 + 8i$, then calculate $\frac{z_1}{z_2}$.

Solution:

$$\frac{z_1}{z_2} = (x_1+iy_1)\left(\frac{x_2}{x_2^2+y_2^2} - i\frac{y_2}{x_2^2+y_2^2}\right)$$

$$= \left(\frac{x_1x_2+y_1y_2}{x_2^2+y_2^2}, \frac{y_1x_2-x_1y_2}{x_2^2+y_2^2}\right)$$

$$= \left(\frac{(-8)(-3)+(1)(8)}{(-3)^2+(8)^2}, \frac{(1)(-3)-(-8)(8)}{(-3)^2+(8)^2}\right)$$

$$= \left(\frac{32}{(-3)^2+(8)^2}, \frac{61}{(-3)^2+(8)^2}\right)$$

Q6: Find the modulus $|z|$ of the complex numbers $z = -7 + 5i$.

Solution: $|z| = |-7 + 5i| = \sqrt{7^2+5^2} = \sqrt{49+25} = \sqrt{74}$

Q7: Find the distance of the complex number $z_1 = 2 + i$ from $z_2 = -3 + 9i$.

Solution: Distance $= \sqrt{(2-(-3))^2+(1-9)^2} = \sqrt{(5)^2+(-8)^2}$

$= \sqrt{25+64} = \sqrt{89}$

Q8: Show that $|z_1 + z_2| \geq ||z_1| - |z_2||$

Proof

$|z_1| = |z_1 + z_2 - z_2| \leq |z_1+z_2| + |-z_2|$ $\quad \because |z| = |-z|$

$\Rightarrow |z_1| - |z_2| \leq |z_1+z_2|$ — I

Interchanging z_1 & z_2

$|z_2| - |z_1| \leq |z_1+z_2|$



$-\alpha < x < \alpha$

$|x| < \alpha$

$\Rightarrow -(|z_1| - |z_2|) \leq |z_1+z_2|$

$|z_1| - |z_2| \geq -|z_1+z_2|$ — II

$-|z_1+z_2| \leq |z_1| - |z_2| \leq |z_1+z_2|$ — III

PRACTICE QUESTIONS.

By Pin2

MTH632

Lecture No. 3

Modules (8 – 9)

Q1: Verify that $|z|^2 = z\bar{z}$ for $z = -3 + 4i$.

Solution:

$z = -3 + 4i$, so $\bar{z} = -3 + 4i = -3 - 4i$

$z\bar{z} = (-3 + 4i)(-3 - 4i)$

$= (-3)(-3) + (-3)(-4i) + (4i)(-3) + (4i)(-4i)$

$= 9 + 12i - 12i - 16\, i^2$

$= 9 - 16\,(-1)$

$z\bar{z} = 25$

Now $|z| = \sqrt{(-3)^2+(4)^2} = \sqrt{25} = 5$

$|z|^2 = 5^2 = 25$

Hence $|z|^2 = z\bar{z}$

Q2: Verify that $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ for $z_1 = 2 + i$, $z_2 = -3 + 9i$.

Solution:

$\bar{z}_1 + \bar{z}_2 = (2+i) + (-3+9i)$

$= (2-3) + (i+9i)$

$= -1 + 10i$

$= -1 - 10i$

$$\bar{z}_1 + \bar{z}_2 = \overline{(2+i)} + \overline{(-3+9i)}$$

$$= 2 - i + (-3-9i)$$

$$= -1 - 10i$$

$$\text{Hence} \quad \bar{z}_1 + \bar{z}_2 = \overline{z_1 + z_2}$$

Q3: Show that $\mathrm{Re}(z) = \frac{z+\bar{z}}{2}$ and $\mathrm{Im}(z) = \frac{z-\bar{z}}{2i}$.

Solution:

$$z = x + yi, \quad \text{and} \quad \bar{z} = x - yi$$

$$\mathrm{Re}(z) = x, \qquad \mathrm{Im}(z) = y$$

$$\text{Now } z + \bar{z} = (x + yi) + (x - yi)$$

$$= (x + x) + (yi - yi)$$

$$= 2x + 0i$$

$$z + \bar{z} = 2x$$

$$\frac{z + \bar{z}}{2} = x$$

$$\mathrm{Re}(z) = \frac{z + \bar{z}}{2}$$

$$\text{Now } z - \bar{z} = (x + yi) - (x - yi)$$

$$= (x - x) + (yi + yi)$$

$$= 0 + 2yi$$

$$z - \bar{z} = 2yi$$

$$\frac{z - \bar{z}}{2i} = y$$

$$\mathrm{Im}(z) = \frac{z - \bar{z}}{2i}$$

Q4: Show that $\mathrm{Re}(iz) = -\mathrm{Im}(z)$ and $\mathrm{Im}(iz) = \mathrm{Re}(z)$.

Solution:

$$\text{Since } \mathrm{Re}(z) = \frac{z + \bar{z}}{2}$$

$$\text{So } \mathrm{Re}(iz) = \frac{iz + \overline{iz}}{2}$$

$$= \frac{iz - i\bar{z}}{2}$$

$$= i\left(\frac{\bar{z} - z}{2}\right)$$

$$= i.i\left(\frac{\bar{z} - z}{2i}\right)$$

$$\mathrm{Re}(iz) = -\mathrm{Im}(z)$$

$$\text{Since } \mathrm{Im}(z) = \frac{z - \bar{z}}{2i}$$

$$\text{So } \mathrm{Im}(iz) = \frac{iz - \overline{iz}}{2i}$$

$$= \frac{iz + i\bar{z}}{2i}$$

$$= i\left(\frac{z + \bar{z}}{2i}\right)$$

$$= \left(\frac{z + \bar{z}}{2}\right)$$

$$\mathrm{Im}(iz) = \mathrm{Re}(z)$$

Q5: Verify that the triangle inequality $|z_1 + z_2| \le |z_1| + |z_2|$ for $z_1 = -8 + i$, $z_2 = -3 + 8i$.

Solution:

$z_1 = -8 + i, \quad |z_1| = \sqrt{(-8)^2+1^2} = \sqrt{65} = 8.06$

$z_2 = -3 + 8i, \quad |z_2| = \sqrt{(-3)^2+8^2} = \sqrt{73} = 8.5$

$z_1 + z_2 = (-8 + i) + (-3 + 8i) = -11 + 9i$

$|z_1+z_2| = \sqrt{(-11)^2+9^2} = \sqrt{202} = 14.12$

Clearly $|z_1+z_2| < |z_1| + |z_2|$

14.12 < 8.06+8.5

14.12 < 16.56

Q6: Find the Polar representation of the complex numbers $z = -7 + 5i$.

Solution:

We represent $z = (x, y)$ in (r, θ) in polar form.

For $z = x + iy, \quad r = |z| = \sqrt{(-7)^2+5^2} = \sqrt{74}$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{5}{-7}\right) = -\tan^{-1}\left(\frac{5}{7}\right)$$

$z = r(\cos\theta + i\sin\theta)$

$$z = \sqrt{74}\left(\cos\left(-\tan^{-1}\left(\frac{5}{7}\right)\right)+i\sin\left(-\tan^{-1}\left(\frac{5}{7}\right)\right)\right)$$

Q7: Find the Polar representation of the complex numbers $z = (0, 11) = 11i$.

Solution:

We represent $z = (x, y)$ in (r, θ) in polar form.

For $z = x + iy, \quad r = |z| = \sqrt{(0)^2+11^2} = \sqrt{121} = 11$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{11}{0}\right) = \infty, \quad \text{so } \theta = \frac{\pi}{4}$$

$z = r(\cos\theta + i\sin\theta)$

$$z = 11\left(\cos\left(\frac{\pi}{4}\right)+i\sin\left(\frac{\pi}{4}\right)\right)$$

PRACTICE QUESTIONS

By Pin2

MTH632

Lecture No. 4

Modules (10 – 12)

Q1: Express the complex number $-1+i$ in the exponential form.

Solution:

We represent z = (x, y) in (r, θ)in polar form which is further converted to exponential form $z = re^{-i\theta}$.

For z = x + i y, z = -1 + i, so x = -1, y = 1

$r = |z| = \sqrt{(-1)^2+1^2} = \sqrt{2}$

Since x is –ve and y is +ve, so θ is in 2nd quadrant.

$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-1}{1}\right) = -\tan^{-1}(1) = -\frac{\pi}{4} = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$

$z = re^{-i\theta}$

$z = \sqrt{2}e^{-i\frac{3\pi}{4}}$

Q2: Let $z = 8e^{\frac{\pi i}{2}}$, then find z^{-1}.

Solution:

$$z^{-1} = \frac{1}{re^{i\theta}} = \frac{1}{8e^{i\frac{\pi}{2}}} = \frac{1}{8}e^{-i\frac{\pi}{2}}$$

Q3: Let $z = 7e^{\frac{i\pi}{4}}$, then find z^{50}.

Solution:

$z^n = r^n e^{in\theta}$

Here *n = 50, r = 7,* $\theta = \frac{\pi}{4}$

$z^{50} = 7^{50} e^{i50\left(\frac{\pi}{4}\right)} = 7^{50} e^{i\left(\frac{25\pi}{2}\right)}$

Q4: Use De Moivres' formula to simplify $\left[\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)\right]^{15}$.

Solution:

By De Moivres' formula, $(\cos\theta + i\sin\theta)^n = (\cos n\theta + i\sin n\theta)$

So, $\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^{15} = \left(\cos 15\left(\frac{\pi}{3}\right) + i\sin 15\left(\frac{\pi}{3}\right)\right)$

Q5: Let $z_1 = \sqrt{3}\, e^{\frac{5\pi i}{3}}$ *and* $z_2 = 2\, e^{\frac{\pi i}{3}}$, then find Arg ($z_1 + z_2$).

Solution:

Arg ($z_1 + z_2$) =?

$\text{Arg}(z_1) = \theta_1$, $\text{Arg}(z_2) = \theta_2$

$\text{Arg}(z_1 + z_2) = \theta_1 + \theta_2$

Here,

$\text{Arg}(z_1) = \text{Arg}\left(\sqrt{3}\, e^{i\frac{5\pi}{3}}\right) = \frac{5\pi}{3}$

$\text{Arg}(z_2) = \text{Arg}\left(2\, e^{i\frac{\pi}{3}}\right) = \frac{\pi}{3}$

So

$\text{Arg}(z_1 + z_2) = \theta_1 + \theta_2 = \frac{5\pi}{3} + \frac{\pi}{3} = \frac{6\pi}{3} = 2\pi$

PRACTICE QUESTIONS

By Pin2

MTH632

Lecture No. 5

Modules (13 – 14)

Q1: Find square roots of $2i$.

Solution:

Please follow the solved example at **page 12** of **Module 14.**

Q2: Write down the formula of nth roots of complex number z.

Solution:

Please follow the solved example at **page 13** of **Module 13.**

Q3: Find 6th roots of $2i$.

Solution:

Please follow the solved example at **page 12** of **Module 14.**

Q4: Find cube roots of 1.

Solution:

Please follow the solved example at **page 3** of **Module 14.**

Q5: Find all values of $(3i)^{1/3}$.

Solution:

Please follow the solved example at **page 12** of **Module 14.**

PRACTICE QUESTIONS

By Pin2

MTH632

Lecture No. 6

Modules 14-15

Q1: Find 7th roots of $z_0 = 1$.

Solution:

Please follow the solved example at **page 10** of **Module 14.**

Q2: Find all values of $(5i)^{1/3}$.

Solution:

Please follow the solved example at **page 12** of **Module 14.**

Q3: Find roots of the following equation

$$z^2 - (3 + i)z + \frac{1}{4} = 0$$

Solution:

Please follow the solved example at **page 14** of **Module 14.**

PRACTICE QUESTIONS

By Pin2

MTH632

Lecture No. 8

Modules 18-19

Q1: Explain domain and range of a complex valued function with the help of an example.

Solution:

Please follow the solved example at **pages 6, 11 and 12** of **Module 18.**

Q2: Express the function $f(z) = \bar{z}\,\mathrm{Im}(z) + z\,\mathrm{Re}(z)$ in the form $f(z) = u(x, y) + i\,v(x, y)$.

Solution:

Please follow the solved example at **pages 5-8** of **Module 19.**

Q3: Express the function $f(z) = 8(x + y) + i4y^2$ in the formula, involving z and $\bar{z}$.

Solution:

Please follow the solved example at **pages 9-12** of **Module 19.**

Q4: Define polynomial and give reason why
$f(z) = 1 + z + z^2 + z^3 + z^4 + \cdots$ is NOT a polynomial.

Solution:

Please follow the solved example at **pages 28 and 31** of **Module 19.**

PRACTICE QUESTIONS

By Pin2

MTH632

Lecture No. 9

Modules 20

Q1: Show that the function w = f(z) = iz - 3 is one-to-one.

Solution:

Please follow the solved example at **pages 18 and 19** of **Module 20.**

Q2: Show whether the function w = f(z) = z^2 is one-to-one or not.

Solution:

Please follow the solved example at **pages 21 and 22** of **Module 20.**

Q3: Show that the function f(z) = -z maps the line y = -x + 1 to the line v = -u - 1.

Solution:

Please follow the solved example at **pages 30-36** of **Module 20.**

SOLVED EXERCISE MTH-632

Complex Analysis & Differential Geometry

Instructor

Dr. Sohail Iqbal

Midterm's Syllabus Solved Exercises

Book Name: Brown-Churchill-Complex Variables and Application 8th edition

Chapter 1 Section 2,3,4,8,10,

Chapter 2 Section 12,18,20,**23**,25,26

Section 2 ***Basic Algebraic Operations***

Question No.1: Verify that

a): $\left(\sqrt{2}-i\right)-i\left(1-\sqrt{2}\right)=-2i$

proof:

$$\begin{aligned} L.H.S &= \left(\sqrt{2}-i\right)-i\left(1-\sqrt{2}i\right) \\ &= \sqrt{2}-i-i+\sqrt{2}i^2 \\ &= \sqrt{2}-2i+\sqrt{2}\left(-1\right) \\ &= \sqrt{2}-2i-\sqrt{2} \\ &= -2i = R.HS \end{aligned}$$

b):

$$
\begin{aligned}
L.H.S &= (2,-3)(-2,1) \\
&= (2-3i)(-2+i) \\
&= 2(-2+i)-3i(-2+i) \\
&= -4+2i+6i-3i^2 \\
&= -4+8i-3(-1) \\
&= -4+8i+3 \\
&= -1+8i = R.HS
\end{aligned}
$$

c):

$$
\begin{aligned}
L.H.S &= (3,1)(3,-1)\left(\frac{1}{5},\frac{1}{10}\right) \\
&= (3+i)(3-i)\left(\frac{1}{5}+\frac{1}{10}i\right) \\
&= \left(3(3-i)+i(3-i)\right)\left(\frac{1}{5}+\frac{1}{10}i\right) \\
&= \left(9-3i+3i-i^2\right)\left(\frac{1}{5}+\frac{1}{10}i\right) \\
&= \left(9-(-1)\right)\left(\frac{1}{5}+\frac{1}{10}i\right) \\
&= 10\left(\frac{1}{5}+\frac{1}{10}i\right) \\
&= 2+i \\
&= (2,1) = R.HS
\end{aligned}
$$

Question No.2:

a): Show that *Re(iz)=-ImgZ;*

Proof:

$$
\begin{aligned}
L.H.S &= \mathrm{Re}(iz) \\
&= \mathrm{Re}\left(i\left(x+iy\right)\right) \\
&= \mathrm{Re}\left(ix+i^2y\right) \\
&= \mathrm{Re}(ix-y) \\
&= -y \\
&= -\mathrm{Im}(z) = R.H.S
\end{aligned}
$$

b): Show that Img(iz)=Re z;

Proof:

$$\begin{aligned} L.H.S &= \mathrm{Im}(iz) \\ &= \mathrm{Im}\left(i\left(x+iy\right)\right) \\ &= \mathrm{Im}\left(ix+i^2y\right) \\ &= \mathrm{Im}(ix-y) \\ &= \mathrm{Re}(z) = R.H.S \end{aligned}$$

Question No.3: Show that $(1+z)^2 = 1+2z+z^2$

Ans:

Question No.4: verify that each of the two numbers $z = 1 \pm i$ satisfies the equation $z^2 - 2z + 2 = 0$

Proof:

a): when $z = 1+i$ then

$$\begin{aligned} L.H.S &= z^2 - 2z + 2 \\ &= (1+i)^2 - 2(1+i) + 2 \\ &= (1)^2 + (i)^2 + 2(1)(i) - 2 - 2i + 2 \\ &= 1 + (-1) + 2i - 2 - 2i + 2 \\ &= 0 = R.H.S \end{aligned}$$

b): when $z = 1-i$ then

$$\begin{aligned} L.H.S &= z^2 - 2z + 2 \\ &= (1-i)^2 - 2(1-i) + 2 \\ &= (1)^2 + (i)^2 - 2(1)(i) - 2 + 2i + 2 \\ &= 1 + (-1) - 2i - 2 + 2i + 2 \\ &= 0 = R.H.S \end{aligned}$$

Question No.5: Prove that multiplication of complex numbers is commutative.

Proof:

Let z_1 and z_2 be two complex numbers such that $z_1 = x_1 + iy_1$ *and* $z_2 = x_2 + iy_2$ where $x_1, x_2, y_1, y_2 \in \mathbb{R}$

$$\begin{aligned} z_1 z_2 &= (x_1 + iy_1)(x_2 + iy_2) \\ &= (x_1)(x_2 + iy_2) + (iy_1)(x_2 + iy_2) \\ &= (x_1)(x_2) + (x_1)(iy_2) + (iy_1)(x_2) + (iy_1)(iy_2) \\ &= (x_1)(x_2) + i(x_1)(y_2) + i(y_1)(x_2) + (i)(i)(y_1)(y_2) \\ &= (x_2)(x_1) + i(y_2)(x_1) + i(x_2)(y_1) + (i)(i)(y_2)(y_1) \end{aligned}$$

As real numbers are commutative

$$\begin{aligned} &= (x_2)(x_1) + i(x_2)(y_1) + i(y_2)(x_1) + (i)(i)(y_2)(y_1) \\ &= (x_2)(x_1 + iy_1) + (iy_2)(x_1 + iy_1) \\ &= (x_2 + iy_2).(x_1 + iy_1) \\ &= z_2 z_1 \end{aligned}$$

Hence complex numbers are commutative.

Question No.6: Verify the associative law for addition of complex numbers

Proof:

Let z_1, z_2 and z_3 be three complex numbers such that

$z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$ *and* $z_3 = x_3 + iy_3$ where $x_1, x_2, x_3, y_1, y_2, y_3 \in \mathbb{R}$

$$\begin{aligned} (z_1 + z_2) + z_3 &= \big((x_1 + iy_1) + (x_2 + iy_2)\big) + (x_3 + iy_3) \\ &= \big((x_1 + x_2) + i(y_1 + y_2)\big) + (x_3 + iy_3) \\ &= \big((x_2 + x_1) + i(y_2 + y_1)\big) + (x_3 + iy_3) \end{aligned}$$

As real numbers are commutative

$$
\begin{aligned}
&= (x_1 + x_2) + x_3 + i(y_1 + y_2) + iy_3 \\
&= x_1 + (x_2 + x_3) + i((y_1 + y_2) + y_3) \\
&= x_1 + (x_2 + x_3) + i(y_1 + (y_2 + y_3)) \\
&= x_1 + (x_2 + x_3) + iy_1 + i(y_2 + y_3) \\
&= x_1 + iy_1 + (x_2 + x_3) + i(y_2 + y_3) \\
&= z_1 + (z_2 + z_3)
\end{aligned}
$$

Question No.7:

Ans:

Question No.8:

a): Write $(x, y) + (u, v) = (x, y)$ and point out how it follows that the complex number 0=(0,0) is unique as an additive identity.

Ans:

$$
\begin{aligned}
&(x, y) + (u, v) = (x, y) \\
&(x + iy) + (u + iv) = (x + iy) \\
&(x + u) + i(y + v) = (x + iy)
\end{aligned}
$$

Comparing the real and imaginary numbers on both sides,

$$x + u = x \quad and \quad y + v = y$$

$$
\begin{aligned}
& x + u = x \quad and \quad y + v = y \\
\Rightarrow \quad & u = x - x \quad and \quad v = y - y \\
& u = 0 \qquad and \quad v = 0
\end{aligned}
$$

Hence $(u, v) = (0, 0) = 0$

Suppose there is another additive $(c + id)$ identity then

$$
\begin{aligned}
&(x, y) + (c, d) = (x, y) \\
&(x + iy) + (c + i) = (x + iy) \\
&(x + c) + i(y + d) = (x + iy)
\end{aligned}
$$

$$
\begin{aligned}
& x + c = x \quad and \quad y + d = y \\
\Rightarrow \quad & c = x - x \quad and \quad d = y - y \\
& c = 0 \qquad and \quad d = 0
\end{aligned}
$$

Hence $(c,d)=(0,0)=0=(u,v)$ proved that additive identity is a unique number.

Question No.9: use -1=(-1,0) and z=(x,y) to show that (-1)z=-z

Ans:

Given that $-1=(-1,0)=-1+i0$ and $z=(x,y)=x+iy$

$$\begin{aligned} L.H.S &= (-1)z \\ &= (-1,0)(x,y) \\ &= (-1+i0)(x+iy) \\ &= (-1)(x+iy)+i0(x+iy) \\ &= -x-iy+i0-0 \\ &= -(x+iy) \\ &= -z = R.H.S \end{aligned}$$

Question No.10: use $i=(0,1)$ *and* $y=(y,0)$ to verify that –(iy)=(-i)y. Thus show that additive inverse of a complex number z=x+iy can be written –z=-x-iy without ambiguity.

Proof:

$$i=(0,1) \text{ and } y=(y,0)$$

$$\begin{aligned} L.H.S &= -(iy) \\ &= -\big((0,1)(y,0)\big) \\ &= -\big((0+i)(y+i0)\big) \\ &= -\big((0)(y+i0)+(i)(y+i0)\big) \\ &= -\big(0+i0+iy-0\big) \\ &= -\big((0-0)+(i)(y+0)\big) \\ &= (-i)y = R.H.S \end{aligned}$$

Additive inverse gives the zero number in complex numbers

If $z=x+iy$ then let $w=u+iv$ be the additive inverse of the z , so

$$\begin{aligned} z+w &= 0+i0 \\ (x+iy)+(u+iv) &= 0+i0 \\ (x+u)+i(y+v) &= 0+i0 \end{aligned}$$

Comparing real and imaginary parts

$$x+u=0 \quad and \quad y+v=0$$
$$u=-x \quad and \quad v=-y$$

As

$$w=u+iv$$
$$=-x-iy$$
$$=-(x+iy)$$
$$=-z$$

Hence additive inverse of a complex number z=x+iy can be written –z=-x-iy

Question No.11:solve the equation $z^2+z+1=0$ *for* $z=(x,y)$ by writing

$$(x,y)(x,y)+(x,y)+(1,0)=(0,0)$$

Ans:

$$z^2+z+1=(x,y)(x,y)+(x,y)+(1,0)$$
$$=(x^2-y^2,yx+xy)+(x,y)+(1,0)$$
$$=(x^2-y^2,yx+xy)+(x,y)+(1,0)$$
$$=(x^2-y^2+x+1,yx+xy+y+0)$$

Given $z^2+z+1=0$

Comparing real and imaginary parts

$$(x^2-y^2+x+1,yx+xy+y+0)=(0,0)$$

The real part is $x^2-y^2+x+1=0$ …..……..1 and

The imaginary part is $yx+xy+y+0=0$ ……..…2

Solving equation no. 2

$$2xy+y=0$$
$$y(2x+1)=0$$
$$y=0 \quad or \quad 2x+1=0$$
$$x=\frac{-1}{2}$$

When y=0 then this equation does not satisfied.

Put this value of x in equation 1

$$\left(\frac{-1}{2}\right)^2 - y^2 + \left(\frac{-1}{2}\right) + 1 = 0$$

$$\frac{1}{4} - y^2 - \frac{1}{2} + 1 = 0$$

$$\frac{1-2+4}{4} - y^2 = 0$$

$$\frac{3}{4} - y^2 = 0$$

$$\sqrt{y^2} = \sqrt{\frac{3}{4}}$$

$$y = \pm\frac{\sqrt{3}}{4}$$

Hence $x = \frac{-1}{2}$ and $y = \pm\frac{\sqrt{3}}{4}$

$$z = (x, y) = \left(-\frac{1}{2}, \pm\frac{\sqrt{3}}{4}\right)$$

When y=0 then this equation does not satisfied.

Complex Numbers: Trig Identities: 1

De Moivre's Theorem states that for whole number n,

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta.$$

We can use this fact to derive certain trig identites: an example of a use of complex numbers to do real calculations that would oth we express $\cos 5 \theta$ in terms of powers of $\cos \theta. $

$$\begin{aligned}\cos 5\theta + i\sin 5\theta &= (\cos\theta + i\sin\theta)^5 \\ &= \cos^5\theta + 5i\cos^4\theta\sin\theta + 10i^2\cos^3\theta\sin^2\theta + \\ &\quad 10i^3\cos^2\theta\sin^3\theta + 5i^4\cos\theta\sin^4\theta + i^5\sin^5\theta \\ &= [\cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta] + \\ &\quad i[5\cos^4\theta\sin\theta - 10\cos^2\theta\sin^3\theta + \sin^5\theta].\end{aligned}$$

Comparing real parts,

$$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta.$$

Using the substitution $\sin^2\theta = 1 - \cos^2\theta$,

$$\begin{aligned}\cos 5\theta &= \cos^5\theta - 10\cos^3\theta(1 - \cos^2\theta) + 5\cos\theta(1 - \cos^2\theta)^2 \\ &= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta.\end{aligned}$$

Section No. 3

Solved Exercise

1. Reduce each of these quantities to a real number.

$$\frac{1+2i}{3-4i}+\frac{2-i}{5i}=\left(\frac{1+2i}{3-4i}\right).\left(\frac{3+4i}{3+4i}\right)+\frac{(2-i)i}{(5i)i}$$

$$=\frac{3+4i+6i-8}{(3)^2-(4i)^2}+\frac{2i-i^2}{5i^2}$$

$$=\frac{-5+10i}{9-16(-1)}+\frac{2i-(-1)}{5(-1)}$$

$$=\frac{-5+10i}{25}-\frac{1+2i}{5}$$

$$=\frac{(-5+10i)-5(1+2i)}{25}$$

$$=\frac{-5+10i-5-10i}{25}=\frac{-10}{25}=-\frac{2}{5}$$

b):

$$\frac{5i}{(1-i)(2-i)(3-i)}=\frac{5i(1+i)(2+i)}{\left((1)^2-(i)^2\right)\left((2)^2-(i)^2\right)(3-i)}$$

$$=\frac{5i(2+i+2i+i^2)}{\left(1-(-1)\right)\left(4-(-1)\right)(3-i)}$$

$$=\frac{5i(2+3i-1)}{\left(1+1\right)\left(4+1\right)(3-i)}$$

$$=\frac{5i(1+3i)}{10(3-i)}$$

$$=\frac{i+3i^2}{2(3-i)}$$

$$=\frac{i-3}{2(3-i)}$$

$$=\frac{-(3-i)}{2(3-i)}=-\frac{1}{2}$$

c): $(1-i)^4$ =?

$$(1-i)^2 = (1)^2 + (i)^2 - 2(1)(i)$$
$$= 1\text{-}1\text{-}2i = 2i$$
$$(1-i)^4 = (1-i)^2(1-i)^2$$
$$= (2i)(2i)$$
$$= 4i^2 = 4(-1) = -4$$

2. Show that

$$L.H.S = \frac{1}{1/z} = \frac{1}{z^{-1}} \qquad \because \frac{1}{z} = z^{-1}$$
$$= \frac{1}{z^{-1}} \cdot \frac{z}{z} = \frac{1.z}{z^{-1}.z}$$
$$= \frac{z}{1} = z = R.H.S$$

3. use the associative and commutative laws for multiplication to show that

$$L.H.S = (z_1 z_2)(z_3 z_4)$$
$$= (z_2 z_1)(z_3 z_4) \qquad \because (z_1 z_2) = (z_2 z_1) \quad \textit{commutative law}$$
$$= (z_2)(z_1 z_3)(z_4) \qquad \because (z_1)(z_3 z_4) = (z_1 z_3)(z_4) \quad \textit{associative law}$$
$$= (z_1 z_3)(z_2)(z_4) \qquad \because (z_2)(z_1 z_3) = (z_1 z_3)(z_2) \quad \textit{commutative law}$$
$$= (z_1 z_3)(z_2 z_4) \qquad \because (z_2)(z_4) = (z_2 z_4)$$
$$= R.H.S$$

5. Drive expression

$$\frac{z_1}{z_2} = \frac{x_1 x_2 - y_1 y_2}{x_2^2 + y_2^2} + i\frac{y_1 x_2 - x_1 y_2}{x_2^2 + y_2^2} \quad (z_2 \neq 0)$$

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ where $(z_2 \neq 0)$

$$L.H.S = \frac{z_1}{z_2}$$

$$= \frac{x_1 + iy_1}{x_2 + iy_2} \cdot \frac{x_2 - iy_2}{x_2 - iy_2}$$

$$= \frac{x_1x_2 + i^2y_1y_2 + iy_1x_2 - ix_1y_2}{(x_2 + iy_2)(x_2 - iy_2)}$$

$$= \frac{x_1x_2 + (-1)y_1y_2 + i(y_1x_2 - x_1y_2)}{(x_2)^2 - (iy_2)^2}$$

$$= \frac{x_1x_2 - y_1y_2 + i(y_1x_2 - x_1y_2)}{x_2^{\,2} - (-1)\,y_2^{\,2}}$$

$$= \frac{(x_1x_2 - y_1y_2) + i(y_1x_2 - x_1y_2)}{x_2^{\,2} + y_2^{\,2}}$$

$$= \frac{x_1x_2 - y_1y_2}{x_2^2 + y_2^2} + i\frac{y_1x_2 - x_1y_2}{x_2^2 + y_2^2} = R.H.S$$

6. Drive the identity

$$\left(\frac{z_1}{z_3}\right)\left(\frac{z_2}{z_4}\right) = \frac{z_1z_2}{z_3z_4}$$

$$L.H.S = \left(\frac{z_1}{z_3}\right)\left(\frac{z_2}{z_4}\right)$$

$$= (z_1)\left(\frac{1}{z_3}\right)(z_2)\left(\frac{1}{z_4}\right) \qquad \because \left(\frac{z_1}{z_2}\right) = (z_1)\left(\frac{1}{z_2}\right)$$

$$= (z_1)\left(z_3^{-1}\right)(z_2)\left(z_4^{-1}\right) \qquad \because \left(z_3^{-1}\right)(z_2) = (z_2)\left(z_3^{-1}\right) \text{ commutative law}$$

$$= (z_1)(z_2)\left(z_3^{-1}\right)\left(z_4^{-1}\right)$$

$$= (z_1z_2)(z_3)^{-1}(z_4)^{-1}$$

$$= (z_1z_2)(z_3)^{-1}(z_4)^{-1}$$

$$= (z_1z_2)(z_3z_4)^{-1}$$

$$= \frac{z_1z_2}{z_3z_4} = R.H.S$$

7. Use the identity to drive the cancellation law,

$$\frac{z_1 z}{z_2 z} = \frac{z_1}{z_2}$$

$$L.H.S. = \frac{z_1 z}{z_2 z}$$

$$= \frac{z_1 z}{z_2 z} \cdot \frac{z^{-1}}{z^{-1}}$$ multiplying numerator and denomerator by z^{-1}

$$= \frac{z_1}{z_2} \frac{z z^{-1}}{z z^{-1}}$$ associative law

$$= \frac{z_1}{z_2} \frac{1}{1}$$ $\because z z^{-1} = 1$

$$= \frac{z_1}{z_2} = R.H.S$$

Section 8

Exponential form

1. Find the principal argument Arg z when

a): $z = \dfrac{i}{-2-2i}$

Answer:

Definition: (principle value of arg of z):-

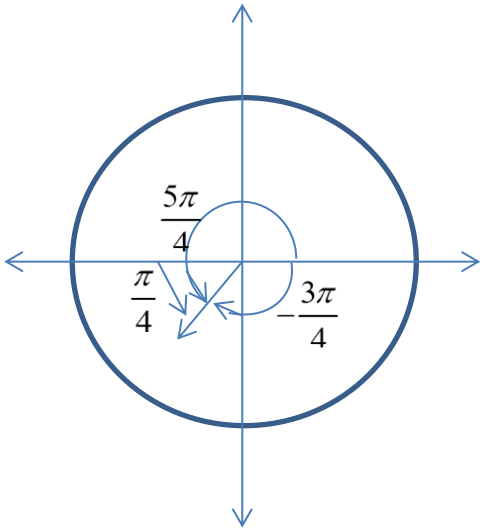
Let $z \neq 0$ be a complex number. Then principle value of argument of z, denoted by $Arg\ z$ is a unique value of θ such that

$$z = r(\cos\theta + i\sin\theta) \qquad -\pi < \theta \leq \pi$$

Principle value of argument of z is also referred as **"The argument of z"**

$$\arg z = \text{Arg}\, z + 2n\pi \qquad n = 0, \pm 1, \pm 2, \pm 3, -----$$

$$z = \frac{i}{-2-2i}$$
$$= \frac{i}{-2(1+i)} \cdot \frac{(1-i)}{(1-i)}$$
$$= \frac{i-i^2}{-2\left((1)^2-(i)^2\right)}$$
$$= \frac{i-(-1)}{-2\left(1-(-1)\right)}$$
$$= \frac{1+i}{-2\left(1+1\right)}$$
$$= \frac{1+i}{-4} = -\frac{1}{4} - i\frac{1}{4}$$



$$z = x + iy = -\frac{1}{4} - i\frac{1}{4}$$
$$x = -\frac{1}{4} \quad , \quad y = -\frac{1}{4}$$
$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-1/4}{-1/4}\right) = \tan^{-1}(1) = \frac{\pi}{4}$$

As x and y are negative so angle lies in the 3rd quadrant, so we add π to $\frac{\pi}{4}$.

$$Arg\ z = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$$

this angle in anti-clockwise direction.

For clockwise direction from positive axis the angle is

$$Arg\ z = \frac{5\pi}{4} - 2\pi = -\frac{3\pi}{4}$$

See example *z=-1-i* on page 23 in handouts of MTH 632 written by Fiddling writer.

Second method.

$$z = \frac{i}{-2-2i}$$

$Arg\ z = Arg\ (i) - Arg(-2 - 2i)$

For $Arg(i) = (0+i)$

If $z \neq 0$ *and* $x = 0$, then we use the following rule:

$$Arg\ z = \frac{\pi}{2} \qquad If \quad \text{Im}\ z > 0$$

$$Arg\ z = -\frac{\pi}{2} \qquad If \quad \text{Im}\, z < 0$$

So, for $Arg(i) = (0+i) = \frac{\pi}{2}$

For Arg(-2-2i)=$\tan^{-1}\left(\frac{-2}{-2}\right) = \tan^{-1}(1) = \frac{\pi}{4}$

As x and y are negative so angle lies in the 3rd quadrant, so we add π to $\frac{\pi}{4}$.

$$Arg(-2 - 2i) = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$$

$$Arg\ z = Arg\ (i) - Arg(-2 - 2i)$$

$$Arg\ z = \frac{\pi}{2} - \frac{5\pi}{4}$$

$$Arg\ z = \frac{2\pi - 5\pi}{4} = -\frac{3\pi}{4}$$

b): $z = (\sqrt{3} - i)^6$

$$r = |z| = \sqrt{x^2 + y^2}$$

$$= \sqrt{\left(\sqrt{3}\right)^2 + (-1)^2} = \sqrt{3+1} = 2$$

θ=$\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$

To write in rectangular form.

$$(\sqrt{3}-i)=re^{i\theta}=2e^{i\frac{\pi}{6}}$$

$$z=(\sqrt{3}-i)^6=\left(2e^{i\frac{\pi}{6}}\right)^6$$

$$=2^6e^{i\left(\frac{\pi}{6}\right)6}=64e^{i\pi}$$

$$\theta=\pi$$

Check.(For principal angle we subtract angle $\frac{\pi}{6}$ from π and here we just find the angle so we do not subtract.)

2. show that $\left|e^{i\theta}\right|=1$

Solution:

$$L.H.S=\left|e^{i\theta}\right|=\left|\cos\theta+i\sin\theta\right|$$
$$=\sqrt{(\cos\theta)^2+(\sin\theta)^2}$$
$$=\sqrt{\cos^2\theta+\sin^2\theta}=\sqrt{1}$$
$$=1=R.H.S$$

show that $\overline{e^{i\theta}}=e^{-i\theta}$

Solution:

$$L.H.S=\overline{e^{i\theta}}=\overline{\cos\theta+i\sin\theta}$$
$$=\cos\theta-i\sin\theta=\cos\theta+(-i)\sin\theta$$
$$=e^{-i\theta}=R.H.S$$

4. show that $\left|e^{i\theta}-1\right|=2$

Solution:

$$L.H.S=\left|e^{i\theta}-1\right|$$
$$=\left|\cos\theta+i\sin\theta-1\right| \qquad 0\le\theta\le 2\pi$$
$$when \quad \theta=\pi$$
$$=\left|\cos\pi+i\sin\pi-1\right|$$
$$=\left|-1+0-1\right|$$
$$=\left|-2\right|=2=R.H.S$$

5. By writing the individual factors on the left in exponential form, performing the needed operations and changing back to rectangular coordinates, show that

a): $i(1-\sqrt{3})(\sqrt{3}+i)=2(1+\sqrt{3}i)$

Solution:

Exponential form are

$$i=e^{i\frac{\pi}{2}}=\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}=i$$
$$(1-\sqrt{3}i)=2\cos\frac{\pi}{3}-2i\sin\frac{\pi}{3}=2e^{-i\frac{\pi}{3}}$$
$$(\sqrt{3}+i)=2\cos\frac{\pi}{6}+2i\sin\frac{\pi}{6}=2e^{i\frac{\pi}{6}}$$

$$L.H.S=i(1-\sqrt{3}i)(\sqrt{3}+i)$$
$$=e^{i\frac{\pi}{2}}\left(2e^{-i\frac{\pi}{3}}\right)\left(2e^{i\frac{\pi}{6}}\right)$$
$$=4e^{i\frac{\pi}{2}-i\frac{\pi}{3}+i\frac{\pi}{6}}$$
$$=4e^{i\frac{3\pi-2\pi+\pi}{6}}$$
$$=4e^{i\frac{\pi}{3}}$$
$$=2(2e^{i\frac{\pi}{3}})$$
$$=2(1+\sqrt{3}i)$$
$$=R.H.S$$

b): $$L.H.S = \frac{5i}{2+i} = \frac{5e^{i\frac{\pi}{2}}}{\sqrt{3}e^{i}}$$

8. Prove: $$\exp\left(i\frac{\theta_1+\theta_2}{2}\right)\exp\left(i\frac{\theta_1-\theta_2}{2}\right) = \exp(i\theta_1)$$

Solution:

$$L.H.S = \exp\left(i\frac{\theta_1+\theta_2}{2}\right)\exp\left(i\frac{\theta_1-\theta_2}{2}\right)$$
$$= \left(\cos\left(\frac{\theta_1+\theta_2}{2}\right)+i\sin\left(\frac{\theta_1+\theta_2}{2}\right)\right)\left(\cos\left(\frac{\theta_1-\theta_2}{2}\right)+i\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right)$$
$$= \left(\cos\left(\frac{\theta_1+\theta_2}{2}\right)\cos\left(\frac{\theta_1-\theta_2}{2}\right)-\sin\left(\frac{\theta_1+\theta_2}{2}\right)\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right)$$
$$+\mathrm{i}\left(\sin\left(\frac{\theta_1+\theta_2}{2}\right)\cos\left(\frac{\theta_1-\theta_2}{2}\right)+\cos\left(\frac{\theta_1+\theta_2}{2}\right)\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right)$$
$$= \cos\left(\frac{\theta_1+\theta_2}{2}+\frac{\theta_1-\theta_2}{2}\right)+i\sin\left(\frac{\theta_1+\theta_2}{2}+\frac{\theta_1-\theta_2}{2}\right)$$
$$= \cos(\theta_1)+i\sin(\theta_1)$$
$$= e^{i\theta_1} = \exp(i\theta_1) = R.H.S$$

b): $$L.H.S = \exp\left(i\frac{\theta_1+\theta_2}{2}\right)\overline{\exp\left(i\frac{\theta_1-\theta_2}{2}\right)}$$

$$= \left(\cos\left(\frac{\theta_1+\theta_2}{2}\right)+i\sin\left(\frac{\theta_1+\theta_2}{2}\right)\right)\overline{\left(\cos\left(\frac{\theta_1-\theta_2}{2}\right)+i\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right)}$$
$$= \left(\cos\left(\frac{\theta_1+\theta_2}{2}\right)+i\sin\left(\frac{\theta_1+\theta_2}{2}\right)\right)\left(\cos\left(\frac{\theta_1-\theta_2}{2}\right)-i\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right)$$
$$= \left(\cos\left(\frac{\theta_1+\theta_2}{2}\right)\cos\left(\frac{\theta_1-\theta_2}{2}\right)+\sin\left(\frac{\theta_1+\theta_2}{2}\right)\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right)$$
$$+\mathrm{i}\left(\sin\left(\frac{\theta_1+\theta_2}{2}\right)\cos\left(\frac{\theta_1-\theta_2}{2}\right)-\cos\left(\frac{\theta_1+\theta_2}{2}\right)\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right)$$
$$= \cos\left(\frac{\theta_1+\theta_2}{2}-\frac{\theta_1-\theta_2}{2}\right)+i\sin\left(\frac{\theta_1+\theta_2}{2}-\frac{\theta_1-\theta_2}{2}\right)$$
$$= \cos(\theta_2)+i\sin(\theta_2)$$
$$= e^{i\theta_2} = \exp(i\theta_2) = R.H.S$$

10. use the de Moivre's Formula to drive

a): $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$

Proof:

$$\begin{aligned}
\sin 3\theta &= 3\sin\theta - 4\sin^3\theta \\
\cos 3\theta &= 4\cos^3\theta - 3\cos\theta \\
\tan 3\theta &= \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}
\end{aligned}$$

$$L.H.S = \cos^3\theta - 3\cos\theta\sin^2\theta = \cos^3\theta - 3\cos\theta(1-\cos^2\theta)$$
$$= \cos^3\theta - 3\cos\theta + 3\cos^3\theta$$
$$= 4\cos^3\theta - 3\cos\theta = \cos 3\theta = R.H.S$$

$$(\cos\theta + i\sin\theta)^3 = (cos\theta + isin\theta)(cos\theta + isin\theta)(cos\theta + isin\theta)$$
$$= \left(cos^2\theta - sin^2\theta + i\sin\theta cos\theta + i\cos\theta sin\theta\right)(cos\theta + isin\theta)$$
$$= cos^3\theta + icos^2\theta\sin\theta - sin^2\theta cos\theta - i\sin^3\theta + icos^2\theta\sin\theta - sin^2\theta\cos\theta + i\cos^2\theta\sin\theta - \sin^2\theta\cos\theta$$
$$= cos^3\theta - 3sin^2\theta cos\theta + 3icos^2\theta\sin\theta - i\sin^3\theta$$
$$= cos^3\theta - 3sin^2\theta cos\theta + i(3cos^2\theta\sin\theta - \sin^3\theta)$$

using de Moivre's law

$$\cos 3\theta + i\sin 3\theta = cos^3\theta - 3cos\theta sin^2\theta + i(3cos^2\theta\sin\theta - \sin^3)$$

Hence

$$\cos 3\theta + i\sin 3\theta = cos^3\theta - 3cos\theta sin^2\theta$$

And

$$\sin 3\theta = 3cos^2\theta\sin\theta - \sin^3\theta$$

Section 10

Examples:

1. Find the square root of 2i, Topic 12 time 15 mint

Answer:

We can write square root of 2i as $(2i)^{\frac{1}{2}}$

$$2i = 2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right) = 2\exp\left[i\left(\frac{\pi}{2} + 2k\pi\right)\right] \qquad \text{k=0,1}$$

$$=2e^{i\left(\frac{\pi}{2}+2k\pi\right)}$$

Taking square root

$$(2i)^{\frac{1}{2}} = \left(2e^{i\left(\frac{\pi}{2}+2k\pi\right)}\right)^{\frac{1}{2}} = \sqrt{2}e^{i\left(\frac{\pi}{4}+k\pi\right)}$$

When k=0

$$\sqrt{2}e^{i\left(\frac{\pi}{4}+k\pi\right)} = \sqrt{2}e^{i\left(\frac{\pi}{4}\right)} = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) = 1+i$$

When k=1

$$\sqrt{2}e^{i\left(\frac{\pi}{4}+(1)\pi\right)} = \sqrt{2}e^{i\left(\frac{5\pi}{4}\right)} = \sqrt{2}\left(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\right) = -1-i = -(1+i)$$

So the roots of 2i are $\pm(1+i)$

1. Find the square root of $1-\sqrt{3}i$, Topic 12 time 15 mint

Solution:

$$1-\sqrt{3}i = \cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right) = 2e^{i\left(-\frac{\pi}{6}+2k\pi\right)}$$

Taking square root

$$\left(1-\sqrt{3}i\right)^{\frac{1}{2}} = \left(2e^{i\left(-\frac{\pi}{6}+2k\pi\right)}\right)^{\frac{1}{2}} = \sqrt{2}e^{i\left(-\frac{\pi}{12}+k\pi\right)} \qquad \text{k=0,1}$$

When k=0

$$\left(1-\sqrt{3}i\right)^{\frac{1}{2}} = \sqrt{2}e^{i\left(-\frac{\pi}{12}+k\pi\right)}$$

$$= \sqrt{2}e^{i\left(-\frac{\pi}{12}\right)} = \sqrt{2}\left(\cos\left(-\frac{\pi}{12}\right)+i\sin\left(-\frac{\pi}{12}\right)\right)$$

$$= \sqrt{2}\left(\cos\left(\frac{\pi}{12}\right)-i\sin\left(\frac{\pi}{12}\right)\right) = -\sqrt{2}\left(\frac{\sqrt{3}-i}{2}\right) = -\frac{\sqrt{3}-i}{\sqrt{2}}$$

When k=1

$$\left(1-\sqrt{3}i\right)^{\frac{1}{2}} = \sqrt{2}e^{i\left(-\frac{\pi}{12}+(1)\pi\right)}$$

$$= \sqrt{2}e^{i\left(\frac{11\pi}{12}\right)} = \sqrt{2}\left(\cos\left(\frac{11\pi}{12}\right)+i\sin\left(\frac{11\pi}{12}\right)\right)$$

$$= \sqrt{2}\left(\frac{\sqrt{3}-i}{2}\right) = +\frac{\sqrt{3}-i}{\sqrt{2}}$$

So the roots of $1-\sqrt{3}i$ are $\pm\frac{\sqrt{3}-i}{\sqrt{2}}$

2. $(-16)^{\frac{1}{4}} = (-1\times16)^{\frac{1}{4}} = \left(i^2\times2^4\right)^{\frac{1}{4}} = 2\left(i\right)^{\frac{1}{2}} = 2\sqrt{i}$

We find the square root of i.

Topic 25 examples:

Complex Limits

Nonexistence of limits

Example: Show that the limit

$$\lim_{z\to 0} \frac{z}{\bar{z}}$$

Does not exist.

Solution: First we calculate the limit when z approaches to 0 along the real axis

$$\lim_{z\to 0} \frac{z}{\bar{z}} = \lim_{x\to 0} \frac{x+i0}{x-i0} = \lim_{x\to 0} 1 = 1.$$

If z approaches 0 along the imaginary axis.

$$\lim_{z\to 0} \frac{z}{\bar{z}} = \lim_{y\to 0} \frac{0+iy}{0-iy} = \lim_{x\to 0} -1 = -1.$$

As 1 is not equal to -1 so, limit does not exist.

Complex Limits

Example: Show that

$$\lim_{z\to z_0} \bar{z} = \bar{z}_0$$

Solution:

$\varepsilon > 0$

$|f(z) - \bar{z}_0| < \varepsilon$

$\Leftrightarrow |\bar{z} - \bar{z}_0| < \varepsilon$

$\Leftrightarrow |\overline{z - z_0}| < \varepsilon$

$\Leftrightarrow |z - z_0| < \varepsilon$

$\delta = \varepsilon$

$|z - z_0| < \delta = \varepsilon$

$\Rightarrow |f(z) - \bar{z}_0| < \varepsilon$

$|f(z) - \bar{z}_0| < \varepsilon$ whenever $|z - z_0| < \delta$

Example: Show that

$$\lim_{z \to z_o} \overline{z} = \overline{z_o}$$

Solution:

Let ε be any real non-negative number. $\varepsilon > 0$.

$$\left| f(z) - \overline{z_o} \right| < \varepsilon$$

$$\Leftrightarrow \quad \left| \overline{z} - \overline{z_o} \right| < \varepsilon$$

$$\Leftrightarrow \quad \left| \overline{z - z_o} \right| < \varepsilon$$

$$\Leftrightarrow \quad \left| z - z_o \right| < \varepsilon$$

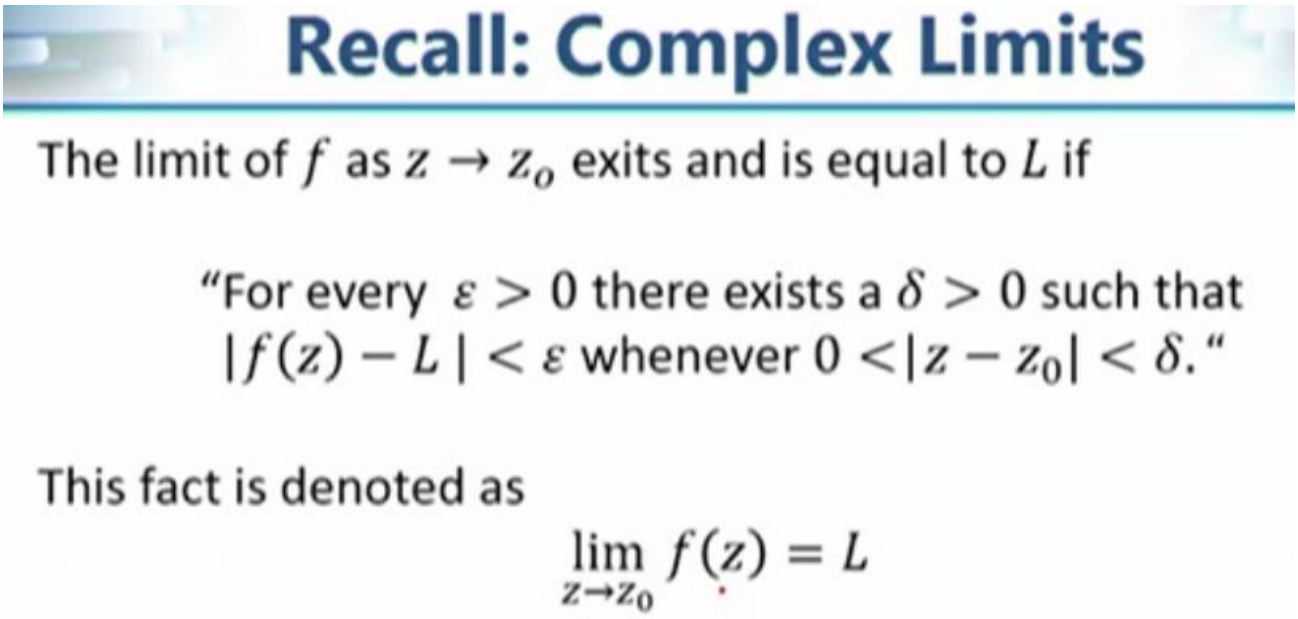
If $\delta = \varepsilon$

$$\left| z - z_o \right| < \delta = \varepsilon$$

$$\Rightarrow \quad \left| f(z) - \overline{z_o} \right| < \varepsilon$$

$\left| f(z) - z_o \right| < \varepsilon$ whenever $\left| z - z_o \right| < \delta s$

Topic 26



Complex Limits

A complex function is of the form

$$f(z) = f(x+iy) = u(x,y) + iv(x,y)$$

We find a relation between the complex limit of $f(z)$ and the real limits of the multivariable real function $u(x,y)$ and $v(x,y)$.

Relation between Real and Complex Limits

Main motivation

Theorem

Let $f(z) = u(x,y) + iv(x,y)$ be a complex function that is defined in some neighborhood of z_0, except perhaps at $z_0 = x_0 + iy_0$. Then

$$\lim_{z \to z_0} f(z) = u_0 + iv_0$$

iff

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0 \text{ and } \lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0 .$$

Example: if $u(x,y) = \dfrac{2x^3}{(x^2+y^2)}$ then show that.

$$\lim_{(x,y)\to(0,0)} u(x,y) = 0$$

Solution:

$x = r\cos\theta,$ y=rsinθ

$$u(r\cos\theta,\ r\sin\theta) = \frac{2r^3\cos^3\theta}{r^2\cos^2\theta + r^2\sin^2\theta} = 2r\cos^3\theta$$

In polar form:

$$|u(x,y)-0|$$
$$\left|2r\cos^3\theta\right| = 2r\left|\cos^3\theta\right| \le 2r$$
$$2r < \varepsilon \quad \Rightarrow \quad r < \frac{\varepsilon}{2}$$

To find $\delta > 0$ such that $|u(x,y)-0| < \varepsilon$

$$|u(x,y)-0| \le 2r < \varepsilon \text{ when ever } \; 0 < \sqrt{(x-0)^2+(y-0)^2} < \delta$$

Exampe: Show that the function

$$u(x,y) = \frac{xy}{x^2+y^2}$$

Does not have a limit as (x,y) approaches (0,0).

Solution:

For y=-x

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = \lim_{x\to 0} \frac{-x^2}{x^2+y^2}$$
$$= \lim_{x\to 0} \frac{-x^2}{x^2+(-x)^2} = \lim_{x\to 0} \frac{-x^2}{2x^2}$$
$$= \lim_{x\to 0} \frac{-1}{2} = -\frac{1}{2}$$

Topic 29

Real Limits of Multivariable functions

Properties of real limits of functions of two variables

Theorem: If $\lim_{(x,y)\to(x_0,y_0)} F(x,y) = A$

and

$$\lim_{(x,y)\to(x_0,y_0)} G(x,y) = B$$

then

1. $\lim_{(x,y)\to(x_0,y_0)} cF(x,y) = cA$, where c is a real constant
2. $\lim_{(x,y)\to(x_0,y_0)} F(x,y) \pm G(x,y) = A \pm B$
3. $\lim_{(x,y)\to(x_0,y_0)} F(x,y).G(x,y) = A.B$
4. $\lim_{(x,y)\to(x_0,y_0)} \frac{F(x,y)}{G(x,y)} = \frac{A}{B}, B \neq 0$

Example:

Calculate the limit

$$\lim_{(x,y)\to(1,2)} 3xy^2 - y$$

Solution:

$$\lim_{(x,y)\to(1,2)} 3xy^2 - y$$

$$= \lim_{(x,y)\to(1,2)} 3xy^2 - \lim_{(x,y)\to(1,2)} y$$

$$= \left(\lim_{(x,y)\to(1,2)} 3x\right)\left(\lim_{(x,y)\to(1,2)} y^2\right) - \left(\lim_{(x,y)\to(1,2)} y\right)$$

$$= 3(1)(2)^2 - 2 = 12 - 2 = 10$$

Example: ***Compute the limit***

$$\lim_{z\to i} z^2 + z + 1$$

Solution:

$$f(z) = z^2 + z + 1$$

$$f(x+iy) = (x+iy)^2 + (x+iy) + 1$$

$$= (x^2 - y^2) - 2ixy + (x+iy) + 1$$

$$= (x^2 - y^2 + x + 1) - i(-2xy + y)$$

$$\lim_{(x,y)\to(0,1)} x^2 - y + x + 1 = -1 + 1 = 0$$

$$\lim_{(x,y)\to(0,1)} -2xy + y = 1$$

$$\Rightarrow \quad \lim_{z\to i} f(z) = i$$

Properties of Complex Limits

Theorem: Suppose that

$$\lim_{z\to z_0} f(z) = A$$

and

$$\lim_{z\to z_0} g(z) = B$$

then

1. $\lim_{z\to z_0} c\,f(z) = cA$, c a complex constant.
2. $\lim_{z\to z_0} (f(z) \pm g(z)) = A \pm B.$
3. $\lim_{z\to z_0} (f(z)\,.\,g(z)) = A.B$
4. $\lim_{z\to z_0} \frac{f(z)}{g(z)} = \frac{A}{B}$ provided $B \neq 0.$

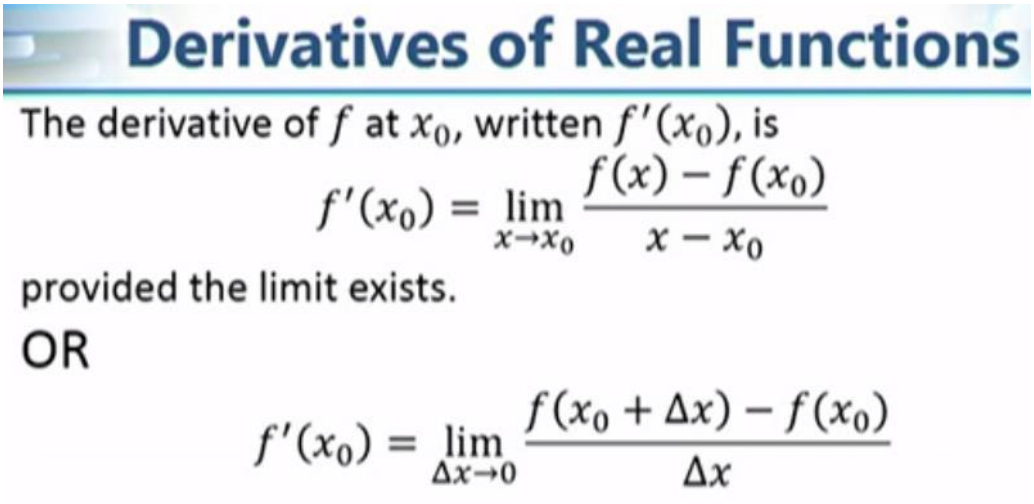
Example:

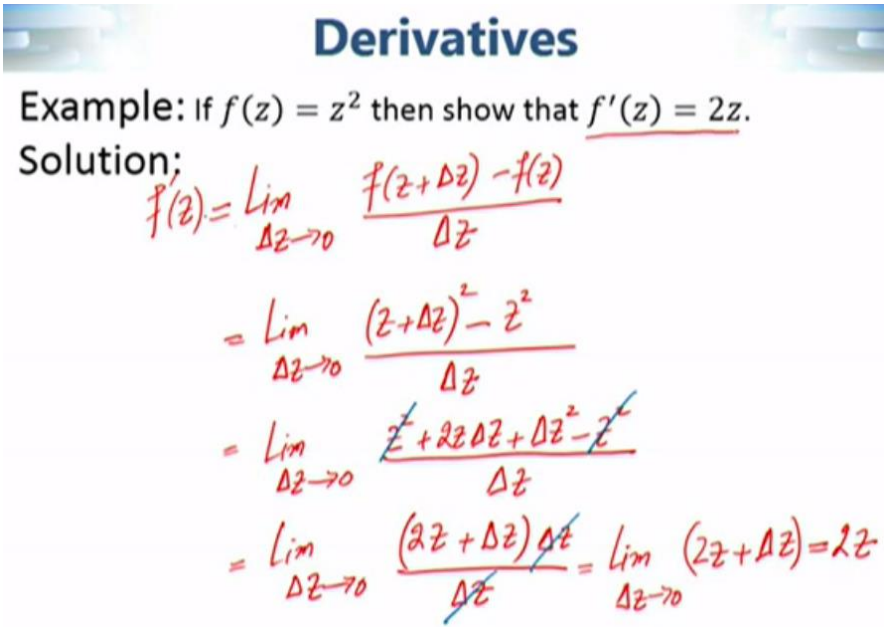
Compute the limit

$$\lim_{z\to i} z^2+z+1$$
$$=\left(\lim_{z\to i} z\right)\left(\lim_{z\to i} z\right)+\left(\lim_{z\to i} z\right)+\left(\lim_{z\to i} 1\right)$$
$$=(i)(i)+i+1$$
$$=-1+i+1=i$$

Topic 34

The mean value theorem for complex numbers does not hold.





Derivatives

Example: Show that the function $f(z) = \mathrm{Re}(z)$ is differentiable nowhere.

Solution: First we approach $z_0 = x_0 + iy_0$ along a line parallel to x −axis. In this case $z = x + iy_0$.

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{\Delta z}$$

$$= \lim_{(x+iy_0) \to (x_0+iy_0)} \frac{\mathrm{Re}(x + iy) - \mathrm{Re}(x_0 + iy_0)}{(x + iy_0) - (x_0 + iy_0)}$$

$$= \lim_{(x+iy_0) \to (x_0+iy_0)} \frac{x - x_0}{x - x_0}$$

$$= 1$$

Derivatives

Solution (cont.): Next we approach z_0 along the line parallel to y −axis by forcing z to be of the form $z = x_0 + iy$.

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{\Delta z}$$

$$= \lim_{(x_0+iy) \to (x_0+iy_0)} \frac{Re(x_0 + iy) - Re(x_0 + iy_0)}{(x_0 + iy) - (x_0 + iy_0)}$$

$$= \lim_{(x+iy_0) \to (x_0+iy_0)} \frac{x_0 - x_0}{i(y - y_0)}$$

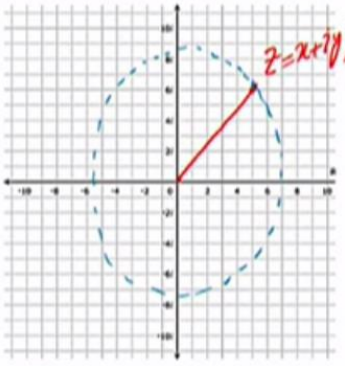
$$= 0$$

Hence the function f(z)=Re(z) is differentiable nowhere.

Derivatives

Example: Consider the real-valued function $f(z) = |z|^2$.
Find points where $f(z)$ is differentiable.

Solution: If $z = x+iy$, $f(z) = f(x+iy) = |x+iy|^2 = x^2+y^2$
Here $f(z)$ is real-valued function.



$f(z) = |z|^2 = z\bar{z}$

Derivatives

Example: Consider the real-valued function $f(z) = |z|^2 = z\bar{z}$
Find points where $f(z)$ is differentiable.

Solution: We know that $f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$

$$f'(z) = \lim_{\Delta z \to 0} \frac{(z+\Delta z)(\overline{z+\Delta z}) - z\bar{z}}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{(z+\Delta z)(\bar{z}+\overline{\Delta z}) - z\bar{z}}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{z\bar{z} + z\overline{\Delta z} + \bar{z}\Delta z + \Delta z\,\overline{\Delta z} - z\bar{z}}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \left(z\frac{\overline{\Delta z}}{\Delta z} + \bar{z} + \overline{\Delta z} \right)$$

Path 1

Derivatives

Example: Consider the real-valued function $f(z) = |z|^2$.
Find points where $f(z)$ is differentiable.

Solution: Path 1: z moves parallel to x-axis

$$\lim_{\Delta z \to 0} \bar{z} + \overline{\Delta z} + z \frac{\overline{\Delta z}}{\Delta z}$$

In this case $\overline{\Delta z} = \Delta z$

$$= \lim_{\Delta z \to 0} \bar{z} + \Delta z + z \frac{\Delta z}{\Delta z}$$

$$= \lim_{\Delta z \to 0} (\bar{z} + \Delta z + z) = \bar{z} + z$$

Derivatives

Example: Consider the real-valued function $f(z) = |z|^2$.
Find points where $f(z)$ is differentiable.

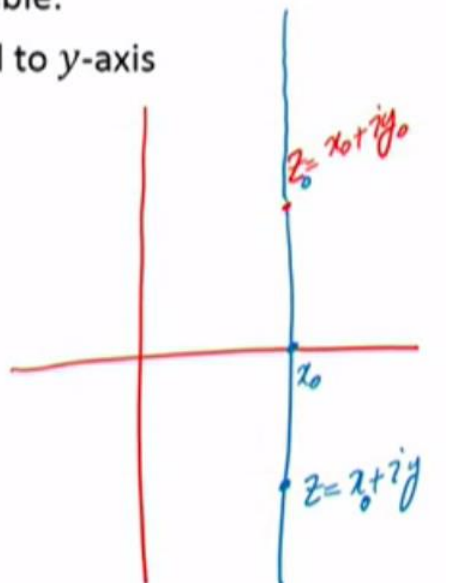
Solution: Path 2: z moves parallel to y-axis

$$\Delta z = z - z_0$$

$$= (x_0 + iy) - (x_0 + iy_0)$$

$$= i(y - y_0)$$

$$\Rightarrow \overline{\Delta z} = -\Delta z$$



Derivatives

Example: Consider the real-valued function $f(z) = |z|^2$.
Find points where $f(z)$ is differentiable.

Solution: Path 2: z moves parallel to y-axis

$$\lim_{\Delta z \to 0} \bar{z} + \overline{\Delta z} + z\frac{\overline{\Delta z}}{\Delta z}$$

In this case $\overline{\Delta z} = -\Delta z$

$$= \lim_{\Delta z \to 0} \bar{z} - \Delta z + z\frac{(-\Delta z)}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \left(\bar{z} - \Delta z - z\right) = \bar{z} - z.$$

Derivatives

Example: Consider the real-valued function $f(z) = |z|^2$.
Find points where $f(z)$ is differentiable.

Solution: Along path 1: parallel to x-axis

$$\lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} = z + \bar{z}$$

along path 2: parallel to y-axis

$$\lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} = \bar{z} - z$$

$$\Rightarrow \quad z + \bar{z} = \bar{z} - z \Rightarrow z = 0.$$

Derivatives

Observations

Example: Consider the real-valued function $f(z) = |z|^2$. Find points where $f(z)$ is differentiable.

① $f(z) = f(x+iy) = (x^2+y^2) + i(0)$

$u(x,y) = x^2+y^2$, $u: \mathbb{R}^2 \to \mathbb{R}$

$v(x,y) = 0$, $v: \mathbb{R}^2 \to \mathbb{R}$

② $f(z)$ is cont but not diff..

Differentiation Formulas

The proofs are very similar to differentiation formulas for real-valued functions

$$\frac{d}{dz}C = 0$$

where C is a constant.

$\frac{d}{dz}(10) = 0$,

$\frac{d}{dz}(i) = 0$

Differentiation Formulas

$$\frac{d}{dz} z^n = n\, z^{n-1}$$

where n is a positive integer.

Example: $\frac{d}{dz}(z^{15})$, $n = 15$

$= 15\, z^{15-1}$

$= 15\, z^{14}$

Differentiation Formulas

$$\frac{d}{dz}[Cf(z)] = C\frac{d}{dz} f(z)$$

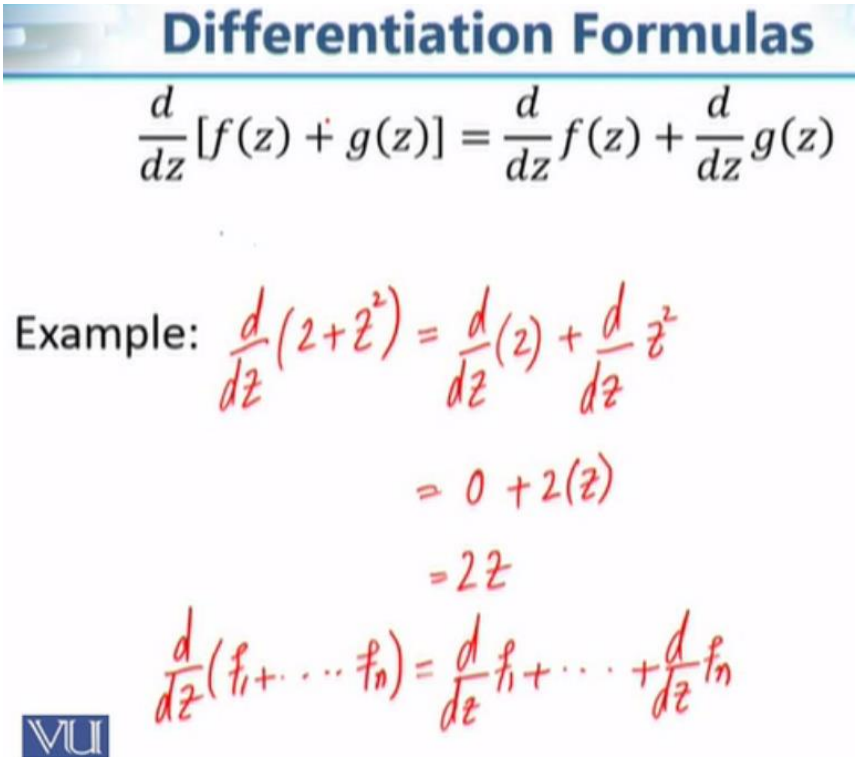
Example:

$\frac{d}{dz}(10 f(z)) = 10 \frac{d}{dz}(f(z))$

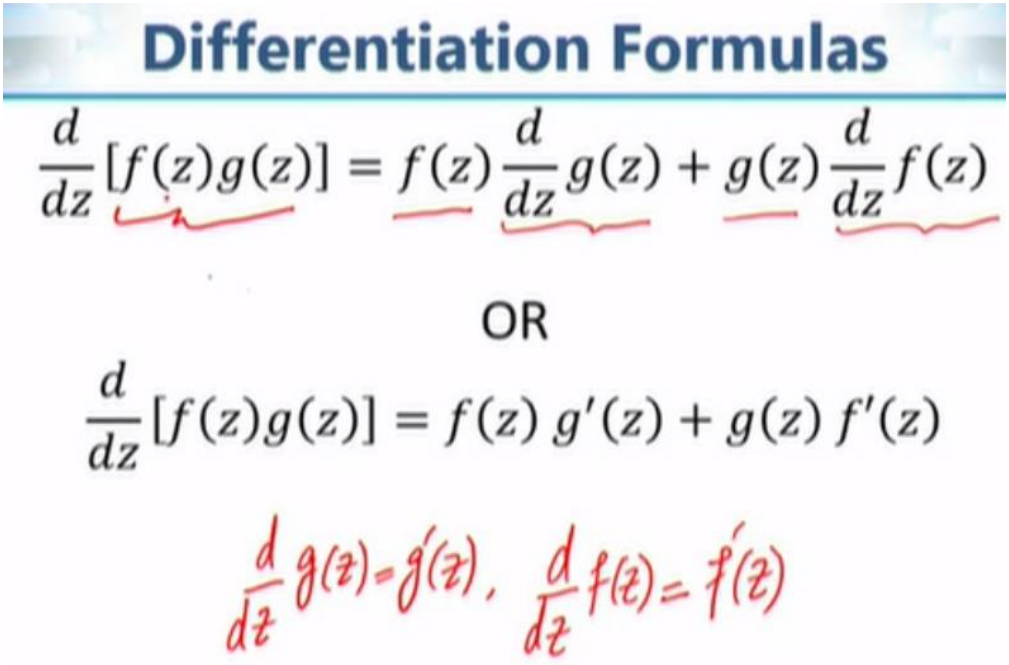
$\frac{d}{dz}(10 z^3) = 10 \frac{d}{dz} z^3 = 10(3z^2)$

$= 30 z^2$.

The Sum Rule



The product Rule



Differentiation Formulas

$$\frac{d}{dz}[f(z)g(z)] = f(z)\,g'(z) + g(z)\,f'(z)$$

Proof: Let $w = f(z)g(z)$

$$w'(z_0) = \lim_{\Delta z \to 0} \frac{w(z+\Delta z) - w(z)}{\Delta z}$$

Now

$$w(z+\Delta z) = f(z+\Delta z)g(z+\Delta z)$$

$$w(z+\Delta z) - w(z) = f(z+\Delta z)g(z+\Delta z) - f(z)\,g(z)$$

Differentiation Formulas

$$w(z+\Delta z) = f(z+\Delta z)g(z+\Delta z)$$

$$w(z+\Delta z) - w(z) = f(z+\Delta z)g(z+\Delta z) - f(z)\,g(z)$$

Adding and subtracting $f(z)g(z+\Delta z)$, we get

$$w(z+\Delta z) - w(z) = f(z+\Delta z)g(z+\Delta z) - f(z)g(z+\Delta z) + f(z)g(z+\Delta z) - f(z)\,g(z)$$

$$w(z+\Delta z) - w(z) = g(z+\Delta z)\,[f(z+\Delta z) - f(z)] + f(z)[g(z+\Delta z) - g(z)]$$

Differentiation Formulas

$$w(z+\Delta z)-w(z)=g(z+\Delta z)\,[f(z+\Delta z)-f(z)] + f(z)[g(z+\Delta z)-g(z)]$$

Dividing by Δz

$$\frac{w(z+\Delta z)-w(z)}{\Delta z}=\frac{g(z+\Delta z)\,[f(z+\Delta z)-f(z)]}{\Delta z} + \frac{f(z)[g(z+\Delta z)-g(z)]}{\Delta z}$$

Taking limit $\Delta z \to 0$

$$\lim_{\Delta z\to 0}\frac{w(z+\Delta z)-w(z)}{\Delta z}=\lim_{\Delta z\to 0} g(z+\Delta z)\,\frac{[f(z+\Delta z)-f(z)]}{\Delta z} + \lim_{\Delta z\to 0} f(z)\,\frac{[g(z+\Delta z)-g(z)]}{\Delta z}$$

Differentiation Formulas

$$\lim_{\Delta z\to 0}\frac{w(z+\Delta z)-w(z)}{\Delta z}=\lim_{\Delta z\to 0} g(z+\Delta z)\,\frac{[f(z+\Delta z)-f(z)]}{\Delta z} + \lim_{\Delta z\to 0} f(z)\,\frac{[g(z+\Delta z)-g(z)]}{\Delta z}$$

$$= \lim_{\Delta z\to 0} g(z+\Delta z)\ \lim_{\Delta z\to 0}\frac{f(z+\Delta z)-f(z)}{\Delta z} + \lim_{\Delta z\to 0} f(z)\ \lim_{\Delta z\to 0}\frac{g(z+\Delta z)-g(z)}{\Delta z}$$

Hence proved the following

$$w'(z)=g(z)f'(z)+f(z)g'(z)$$

Differentiation Formulas

Example: Given $f(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n$ then find

$$\frac{d}{dz} f(z)$$

Solution:

$$\frac{d}{dz}(a_0 + a_1 z + \cdots + a_n z^n) = \frac{d}{dz} a_0 + \frac{d}{dz} a_1 z + \cdots + \frac{d}{dz} a_n z^n$$

$$= a_1 \frac{d}{dz} z + a_2 \frac{d}{dz} z^2 + \cdots + a_n \frac{d}{dz} z^n$$

$$= a_1 + 2a_2 z + 3a_3 z^2 + \cdots + n a_n z^{n-1}$$

Differentiation Formulas

Chain Rule

Chain rule is useful when we want to differentiate composition of two functions.

Given two functions $f(z)$ and $g(z)$ composition of f and g is given by

$$f \circ g(z) = f\big(g(z)\big)$$

Differentiation Formulas

Chain Rule

$$\frac{d}{dz}[f(g(z))] = (f \circ g)'(z) = f'(g(z))\, g'(z)$$

Derivative of outside function

Derivative of inside function

Using Chain Rule

Differentiation Formulas

Example: Given $f(z) = z^3 + i2z^2 + 4$ then find

$$\frac{d}{dz}[f(z)]^3$$

Solution:

$h(z) = z^3, \quad h'(z) = 3z^2$

$(f(z))^3 = h \circ f(z) = h(f(z)) = (f(z))^3$

$$\frac{d}{dz}(f(z))^3 = 3(f(z))^2 f'(z)$$

$$= 3(f(z))^2 \left(3z^2 + 2i(z) + 0\right)$$

The Cauchy-Riemann Equations

Recall

Theorem: Suppose that

$$f(z) = f(x + iy) = u(x, y) + iv(x, y)$$

If $f(z)$ differentiable at a point $z_0 = x_0 + iy_0$ then u and v satisfy **Cauchy-Riemann equations**

$$u_x(x_0, y_0) = v_y(x_0, y_0) \text{ and } u_y(x_0, y_0) = -v_x(x_0, y_0).$$

The Cauchy-Riemann Equations

Example: We know $f(z) = z^2$ is differentiable and $f'(z) = 2z$. Check whether Cauchy-Riemann equations are satisfied by $f(z)$.

Solution: If $z = x+iy$, $f(z) = z^2 = (x+iy)^2 = x^2 - y^2 + i(2xy)$

Here $u(x,y) = x^2 - y^2$, $v(x,y) = 2xy$

Check Cauchy-Riemann equations $u_x = v_y$, $u_y = -v_x$

$u_x = 2x$, $u_y = -2y$, $v_x = 2y$, $v_y = 2x$

$u_x = 2x = v_y$ ✓ $u_y = -2y = -v_x$ ✓

Hence Cauchy Riemann Equations are satisfied.

The Cauchy-Riemann Equations

Example: We Show that $f(z) = \bar{z}$ is nowhere differentiable.

Solution: If $z = x+iy$, $f(z) = f(x+iy) = \overline{(x+iy)} = x+i(-y)$

$\Rightarrow u(x,y) = x$, $v(x,y) = -y$

check Cauchy-Riemann equations $\boxed{u_x = v_y}$, $u_y = -v_x$

$u_x = 1$, $u_y = 0$, $v_x = 0$, $v_y = -1$

$u_x = 1$, $v_y = -1$. If $u_x = v_y$ is satisfied then $1 = -1$

$\Rightarrow$ Cauchy-Riemann are nowhere satisfied.

C-R equations are not satisfied.

The Cauchy-Riemann Equations

Solution: We can rewrite $f(z)$ in the following form

$$f(z) = \begin{cases} \dfrac{(\bar{z})^2}{z} = \dfrac{x^3 - 3xy^2}{x^2+y^2} + i\dfrac{y^3 - 3x^2y}{x^2+y^2} & \text{when } z \neq 0 \\ 0 & \text{when } z = 0 \end{cases}$$

$u(x,y)$

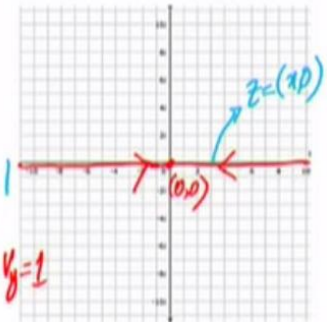
Step1: Checking CR equations

u_x, u_y, v_x, v_y

$u_x(0,0) = \lim_{(x,0)\to(0,0)} \dfrac{u(x,0) - u(0,0)}{x-0}$

$= \lim_{x\to 0} \dfrac{\frac{x^3}{x^2}}{x} = \lim_{x\to 0} \dfrac{x^3}{x^3} = \lim_{x\to 0} 1 = 1$

$\Rightarrow u_x = 1$. Similarly $u_y(0,0) = 0$, $v_x(0,0) = 0$, $v_y = 1$



Uy is equal to vy so it satisfied the equation.

The Cauchy-Riemann Equations

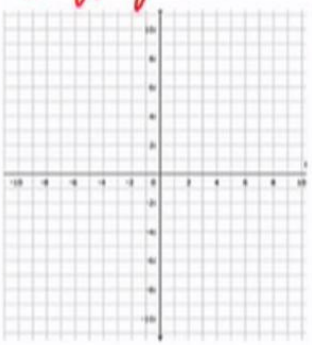
Solution: We can rewrite $f(z)$ in the following form

$$f(z) = \begin{cases} \dfrac{(\bar{z})^2}{z} = \dfrac{x^3 - 3xy^2}{x^2 + y^2} + i\dfrac{y^3 - 3x^2y}{x^2 + y^2} & when\ z \neq 0 \\ 0 & when\ z = 0 \end{cases}$$

Step1: Checking CR equations

$u_x(0,0) = 1,\ u_y(0,0) = 0,\ v_x(0,0) = 0,\ v_y(0,0) = 1$ $\quad u_x = v_y,\ u_y = -v_x$.

Hence Cauchy-Riemann equations are satisfied by $f(z)$



VU

The Cauchy-Riemann Equations

Solution: We can rewrite $f(z)$ in the following form

$$f(z) = \begin{cases} \dfrac{(\bar{z})^2}{z} = \dfrac{x^3 - 3xy^2}{x^2 + y^2} + i\dfrac{y^3 - 3x^2y}{x^2 + y^2} & when\ z \neq 0 \\ 0 & when\ z = 0 \end{cases}$$

Path1: Along x-axis. $z = x + i0$.

$$\lim_{(x,0)\to(0,0)} \frac{f(x+i0) - f(0)}{x - 0}, \quad f(x+i0) = \frac{x^3}{x^2}$$

$$= \lim_{(x,0)\to(0,0)} \frac{\frac{x^3}{x^2}}{x} = \lim_{(x,0)\to(0,0)} \frac{x^3}{x^3} = \lim_{(x,0)\to(0,0)} 1 = 1.$$

The function is not differentiable at the origin.

The Cauchy-Riemann Equations

Cauchy-Riemann Conditions for differentiability

Theorem: Let $f(z) = f(x+iy) = u(x,y) + iv(x,y)$ be defined in some neighborhood of the point $z_0 = x_0 + iy_0$ and suppose that

1. If all the partial derivatives u_x, v_y, u_y and v_x exist everywhere in the neighbourhood.
2. those partial derivatives are continuous at (x_0, y_0) and satisfy Cauchy-Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

Then f is differentiable at z_0, and the derivative $f'(z_0)$ can be computed with either of the following equations

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$

and also

$$f'(z_0) = v_y(x_0, y_0) - iu_y(x_0, y_0).$$

VU

CR Equations in Polar Coordinates

Cauchy-Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

Let us write

$$u_x, u_y, v_x, v_y$$

in terms of

$$u_r, u_\theta, v_r, v_\theta.$$

CR Equations in Polar Coordinates

Chain Rule

Suppose that $u(x, y)$ is a differentiable function of x and y, where $x = x(r, \theta)$ and $y = y(r, \theta)$ are differentiable functions of r and θ. Then

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r}$$

and

$$\frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial \theta}$$

CR Equations in Polar Coordinates

Given $u(x, y)$ and $v(x, y)$ be two component functions of complex valued function

$$f(x + iy) = u(x, y) + iv(x, y)$$

Then

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r}$$

where $x = r\cos\theta, y = r\sin\theta$

$\frac{\partial x}{\partial r} = \cos\theta, \frac{\partial y}{\partial r} = \sin\theta$

$u_r = u_x \cos\theta + u_y \sin\theta$

CR Equations in Polar Coordinates

So we have $u_r = u_x \cos\theta + u_y \sin\theta$; $v_r = v_x \cos\theta + v_y \sin\theta$

$u_\theta = -u_x r\sin\theta + u_y r\cos\theta$; $v_\theta = -v_x r\sin\theta + v_y r\cos\theta$

$u_\theta = -u_x r\sin\theta + u_y r\cos\theta = -r(u_x \sin\theta - u_y \cos\theta)$

$= -r(v_y \sin\theta + v_x \cos\theta)$

$= -rv_r$

$\Rightarrow \boxed{u_\theta = -rv_r,\ ru_r = v_\theta} \Longleftrightarrow u_x = v_y,\ u_y = -v_x$

CR Equations in Polar Coordinates

Theorem: Let the function

$$f(z) = u(r,\theta) + iv(r,\theta)$$

be defined throughout some neighbourhood of a non-zero point $z_0 = r_0\, e^{i\theta_0}$ and suppose that

1. the first order partial derivates of the functions u and v with respect to r and θ exist everywhere in the neighbourhood
2. those partial derivatives are continuous at (r_0, θ_0) and satisfy polar form of Cauchy-Riemann equations

$$ru_r = v_\theta, \qquad u_\theta = -rv_r \quad .$$

Then $f'(z_0)$ exists and its value is

$$f'(z_0) = e^{-i\theta}(u_r + iv_r)$$

where the right side is to be evaluated at (r_0, θ_0)

CR Equations in Polar Coordinates

Example: Show that if f is the principal cube root function given by

$$f(re^{i\theta}) = f(z) = z^{\frac{1}{3}} = r^{\frac{1}{3}}\cos\frac{\theta}{3} + i\, r^{\frac{1}{3}}\sin\frac{\theta}{3},$$

where the domain is restricted to be

$$\{re^{i\theta} : r > 0 \text{ and } -\pi < \theta < \pi\}$$

then the derivative is given by

$$f'(z) = \frac{1}{3z^{\frac{2}{3}}} = \frac{1}{3}r^{-\frac{2}{3}}\cos\frac{\theta}{3} + i\frac{1}{3}r^{-\frac{2}{3}}\sin\frac{\theta}{3}$$

for every point in the domain.

CR Equations in Polar Coordinates

Solution:

$$f(re^{i\theta}) = r^{\frac{1}{3}}\cos\frac{\theta}{3} + i\, r^{\frac{1}{3}}\sin\frac{\theta}{3},$$

be defined throughout some neighbourhood of a non-zero point $z_0 = r_0\, e^{i\theta_0}$ and suppose that

CR Equations in Polar Coordinates

Solution:

$$f(re^{i\theta}) = r^{\frac{1}{3}} \cos\frac{\theta}{3} + i\, r^{\frac{1}{3}} \sin\frac{\theta}{3},$$

the first order partial derivatives of the functions u and v with respect to r and θ exist everywhere in the neighbourhood

$u(r,\theta) = r^{1/3}\cos\frac{\theta}{3}$, $v(r,\theta) = r^{1/3}\sin\frac{\theta}{3}$

$u_r = \frac{1}{3} r^{-2/3}\cos\frac{\theta}{3}$, $v_r = \frac{1}{3} r^{-2/3}\sin\frac{\theta}{3}$

$u_\theta = \frac{-1}{3} r^{1/3}\sin\frac{\theta}{3}$, $v_\theta = \frac{1}{3} r^{1/3}\cos\frac{\theta}{3}$

CR Equations in Polar Coordinates

Solution:

$$f(re^{i\theta}) = r^{\frac{1}{3}} \cos\frac{\theta}{3} + i\, r^{\frac{1}{3}} \sin\frac{\theta}{3},$$

those partial derivatives are continuous at (r_0, θ_0) and satisfy polar form of Cauchy-Riemann equations

$ru_r = v_\theta$ ✓ $u_\theta = -rv_r$

$u_r = \frac{1}{3} r^{-2/3}\cos\frac{\theta}{3}$, $v_\theta = \frac{1}{3} r^{1/3}\cos\frac{\theta}{3} = r\left(\frac{1}{3} r^{-2/3}\cos\frac{\theta}{3}\right)$

$v_\theta = \frac{1}{3} r^{1/3}\cos\frac{\theta}{3}$ $= \frac{1}{3} r^{1/3}\cos\frac{\theta}{3}$

Similarly $u_\theta = -r v_r$ is satisfied.

First C-R equation is satisfied

CR Equations in Polar Coordinates

Solution:

$$f(re^{i\theta}) = r^{\frac{1}{3}}\cos\frac{\theta}{3} + i\, r^{\frac{1}{3}}\sin\frac{\theta}{3},$$

Then $f'(z_0)$ exists and its value is

$$f'(z_0) = e^{-i\theta}(u_r + iv_r)$$

where the right side is to be evaluated at (r_0, θ_0)

$u_r = \frac{1}{3} r^{-2/3}\cos\frac{\theta}{3}$, $v_r = \frac{1}{3} r^{-2/3}\sin\frac{\theta}{3}$

$f'(z_0) = e^{-i\theta_0}\left(\frac{1}{3} r_0^{-2/3}\cos\frac{\theta_0}{3} + i\frac{1}{3} r_0^{-2/3}\sin\frac{\theta_0}{3}\right)$

CR Equations in Polar Coordinates

Solution:

$$f(re^{i\theta}) = r^{\frac{1}{3}}\cos\frac{\theta}{3} + i\, r^{\frac{1}{3}}\sin\frac{\theta}{3},$$

$$\begin{aligned} f'(z) &= e^{-i\theta}\left[\tfrac{1}{3}r^{-\frac{2}{3}}\cos\tfrac{\theta}{3} + i\tfrac{1}{3}r^{-\frac{2}{3}}\sin\tfrac{\theta}{3}\right] \\ &= e^{-i\theta}\left[\tfrac{1}{3}r^{-\frac{2}{3}}e^{\frac{i\theta}{3}}\right] \\ &= \tfrac{1}{3}r^{-\frac{2}{3}}e^{-i\frac{2}{3}} \\ f'(z) &= \frac{1}{3z^{\frac{2}{3}}} \end{aligned}$$

$z^{2/3} = r^{2/3} e^{i\frac{2}{3}\theta}$

$f(z) = z^{1/3}$

The Cauchy-Riemann Equations

Cauchy-Riemann Conditions for differentiability

Theorem: Let $f(z) = f(x + iy) = u(x, y) + iv(x, y)$ be defined in some neighborhood of the point $z_0 = x_0 + iy_0$ and suppose that

1. If all the partial derivatives u_x, v_y, u_y and v_x exist everywhere in the neighbourhood.
2. those partial derivatives are continuous at (x_0, y_0) and satisfy Cauchy-Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

Then f is differentiable at z_0, and the derivative $f'(z_0)$ can be computed with either of the following equations

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$

and also

$$f'(z_0) = v_y(x_0, y_0) - iu_y(x_0, y_0).$$

VU

Exmaples are

The Cauchy-Riemann Equations

Example: Show that the function

$$f(z) = e^{-y}\cos x + i\, e^{-y}\sin x$$

is differentiable for all z and find its derivative.

Solution: Let (x_0, y_0) be an arbitrary point.

defined in some neighborhood of the point $z_0 = x_0 + iy_0$ ✓
and suppose that

$f(z) = f(x+iy) = u(x,y) + iv(x,y)$

$\Rightarrow u(x,y) = e^{-y}\cos x, \quad v(x,y) = e^{-y}\sin x$

$\Rightarrow u(x,y)$ & $v(x,y)$ are defined for all (x,y).

The Cauchy-Riemann Equations

Example: Show that the function

$$f(z) = e^{-y}\cos x + i\, e^{-y}\sin x$$

is differentiable for all z and find its derivative.

Solution:

If all the partial derivatives u_x, v_y, u_y and v_x exist everywhere in the neighbourhood.

$u(x,y) = e^{-y}\cos x$ | $v(x,y) = e^{-y}\sin x$

$u_x(x,y) = -e^{-y}\sin x$ | $v_x(x,y) = e^{-y}\cos x$

$u_y(x,y) = -e^{-y}\cos x$ | $v_y(x,y) = -e^{-y}\sin x$

The Cauchy-Riemann Equations

Example: Show that the function

$$f(z) = e^{-y}\cos x + i\, e^{-y}\sin x$$

is differentiable for all z and find its derivative.

Solution:

those partial derivatives are continuous at (x_0, y_0) and satisfy Cauchy-Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

$u_x = -e^{-y}\sin x$, $v_x = e^{-y}\cos x$

$u_y = -e^{-y}\cos x$, $v_y = -e^{-y}\sin x$

$u_x = -e^{-y}\sin x = v_y$ ✓, $u_y = -e^{-y}\cos x = -v_x$ ✓

The Cauchy-Riemann Equations

Example: Show that the function

$$f(z) = e^{-y}\cos x + i\, e^{-y}\sin x$$

is differentiable for all z and find its derivative.

Solution:

Then f is differentiable at z_0, and the derivative $f'(z_0)$ can be computed with either of the following equations

$$f'(z_0) = u_x(x_0, y_0) + i v_x(x_0, y_0).$$ ✓

and also

$$f'(z_0) = v_y(x_0, y_0) - i u_y(x_0, y_0).$$ ✓

$u_x = -e^{-y}\sin x$, $v_x = e^{-y}\cos x$

$f'(z_0) = -e^{-y_0}\sin x_0 + i\,(e^{-y_0}\cos x_0)$